

Program No: 1

Date: 29/06/2026

Program Title: Create an app that displays name and email id.

MainActivity.java:

```
package com.example.labcycle;
import android.annotation.SuppressLint;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.view.View;
import android.widget.EditText;
import android.widget.Button;
import android.widget.TextView;

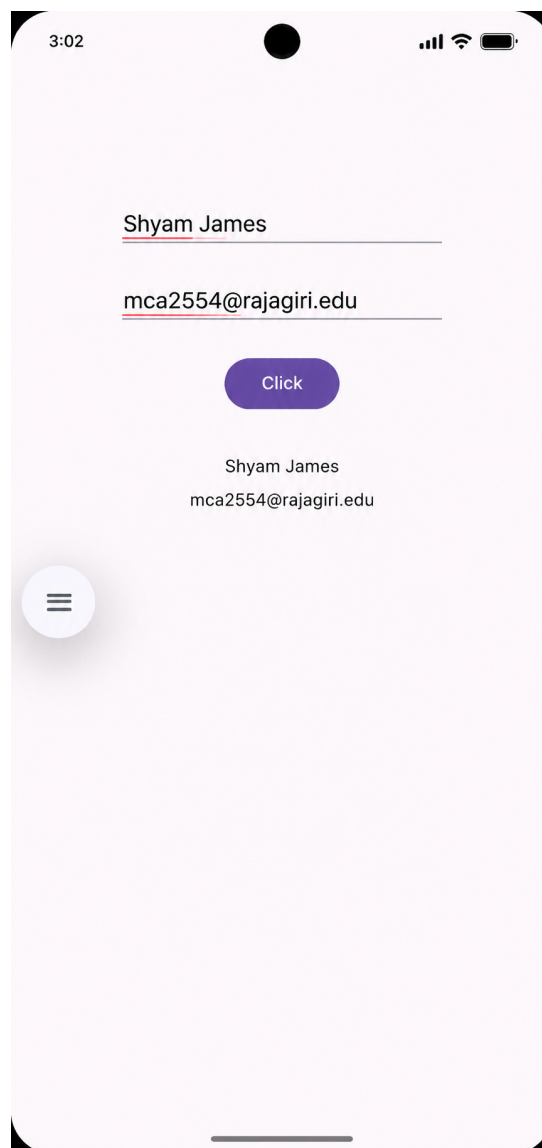
public class MainActivity extends AppCompatActivity {
    EditText name,email;
    Button btn;
    TextView t1,t2;
    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        name = findViewById(R.id.nametxt);
        email = findViewById(R.id.emailTxt);
        btn = findViewById(R.id.btnclick);
        t1 = findViewById(R.id.txtView1);
        t2 = findViewById(R.id.txtView2);
        btn.setOnClickListener(new View.OnClickListener(){
            @Override
            public void onClick(View view) {
                String n1 = name.getText().toString();
                String n2 = email.getText().toString();
                t1.setText(n1);
                t2.setText(n2);
            }
        });
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <EditText
        android:id="@+id/nametxt"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:hint="Name"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="150dp"/>
    <EditText
        android:id="@+id/emailTxt"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:hint="Email"
        app:layout_constraintTop_toBottomOf="@id/nametxt"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp"/>
    <Button
        android:id="@+id/btnclick"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click"
        app:layout_constraintTop_toBottomOf="@id/emailTxt"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="20dp"/>
    <TextView
        android:id="@+id/txtView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name"
        app:layout_constraintTop_toBottomOf="@id/btnclick"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="30dp"/>
```

```
<TextView
    android:id="@+id/txtView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Email"
    app:layout_constraintTop_toBottomOf="@id/txtView1"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="10dp"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 2

Date: 29/06/2026

Program Title: Create an app that takes marks of any three subjects from the user and displays its sum and average.

MainActivity.java:

```
package com.example.labcycle;

import android.annotation.SuppressLint;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.view.View;
import android.widget.EditText;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    EditText m1, m2, m3;
    Button btn;
    TextView t1, t2;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        m1 = findViewById(R.id.mark1);
        m2 = findViewById(R.id.mark2);
        m3 = findViewById(R.id.mark3);
        btn = findViewById(R.id.btncal);
        t1 = findViewById(R.id.textView4);
        t2 = findViewById(R.id.textView5);

        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                int n1 = Integer.parseInt(m1.getText().toString());
                int n2 = Integer.parseInt(m2.getText().toString());
```

```

        int n3 = Integer.parseInt(m3.getText().toString());
        int sums = n1 + n2 + n3;
        int avg = sums / 3;
        t1.setText("Sum is " + sums);
        t2.setText("Average is " + avg);
    }
});
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/mark1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="92dp"
        android:layout_marginBottom="43dp"
        android:ems="10"
        android:inputType="number"
        android:text="mark1"
        app:layout_constraintBottom_toTopOf="@+id/mark2"
        app:layout_constraintEnd_toEndOf="parent" />

    <EditText
        android:id="@+id/mark2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="91dp"
        android:layout_marginBottom="40dp"
        android:ems="10"
        android:inputType="number"
        android:text="mark2"
        app:layout_constraintBottom_toTopOf="@+id/mark3"
        app:layout_constraintEnd_toEndOf="parent" />

    <EditText

```

```

        android:id="@+id/mark3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="85dp"
        android:layout_marginBottom="48dp"
        android:ems="10"
        android:inputType="numberSigned"
        android:text="mark3"
        app:layout_constraintBottom_toTopOf="@+id/btnncal"
        app:layout_constraintEnd_toEndOf="parent" />

<Button
    android:id="@+id/btnncal"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="182dp"
    android:layout_marginBottom="49dp"
    android:text="Button"
    app:layout_constraintBottom_toTopOf="@+id/textView4"
    app:layout_constraintEnd_toEndOf="parent" />

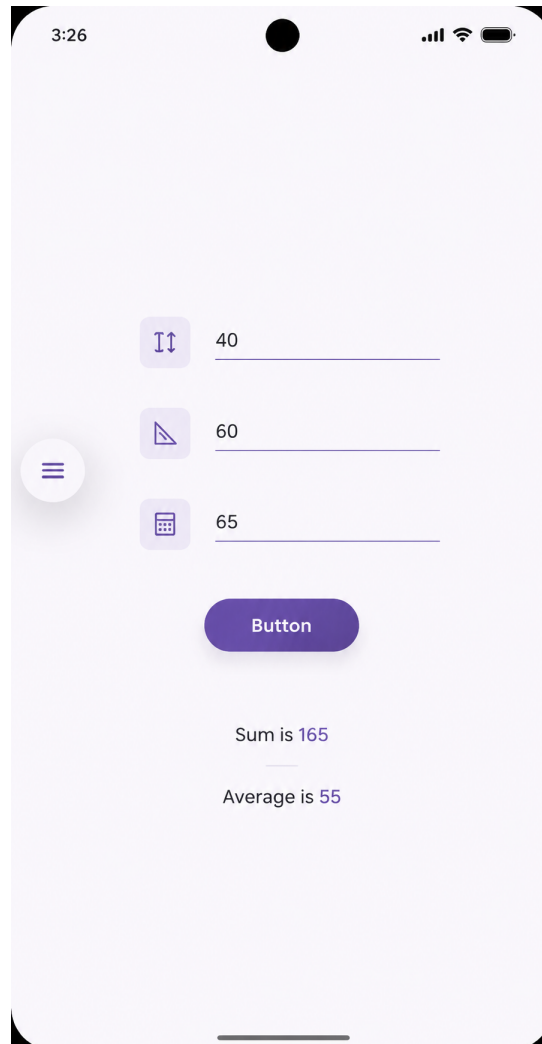
<TextView
    android:id="@+id/textView4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="202dp"
    android:layout_marginBottom="160dp"
    android:text="TextView"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

<TextView
    android:id="@+id/textView5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="200dp"
    android:layout_marginBottom="120dp"
    android:text="TextView"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 3

Date: 29/06/2026

Program Title: Create suitable apps to design the different layouts of the android-studio.

MainActivity.java:

```
package com.example.labcycle;

import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    Button button1, button2, button3;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        button1 = findViewById(R.id.button1);
        button2 = findViewById(R.id.button2);
        button3 = findViewById(R.id.button3);

        button1.setOnClickListener(v ->
            Toast.makeText(this, "LinearLayout Button 1",
                Toast.LENGTH_SHORT).show());
        button2.setOnClickListener(v ->
            Toast.makeText(this, "LinearLayout Button 2",
                Toast.LENGTH_SHORT).show());
        button3.setOnClickListener(v ->
            Toast.makeText(this, "RelativeLayout Button",
                Toast.LENGTH_SHORT).show());
    }
}
```

activity__main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
```



```

        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

<!-- LinearLayout -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Linear Layout"
    android:textStyle="bold"
    android:textSize="18sp"/>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="20dp">

    <Button
        android:id="@+id/button1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Button 1"/>

    <Button
        android:id="@+id/button2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Button 2"/>
</LinearLayout>

<!-- RelativeLayout -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Relative Layout"
    android:textStyle="bold"
    android:textSize="18sp"/>

<RelativeLayout
    android:layout_width="match_parent"

```

```

        android:layout_height="120dp"
        android:background="#DDDDDD"
        android:layout_marginBottom="20dp">

        <TextView
            android:id="@+id/text1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Top Left"/>

        <Button
            android:id="@+id/button3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Bottom Right"
            android:layout_alignParentBottom="true"
            android:layout_alignParentEnd="true"/>
    </RelativeLayout>

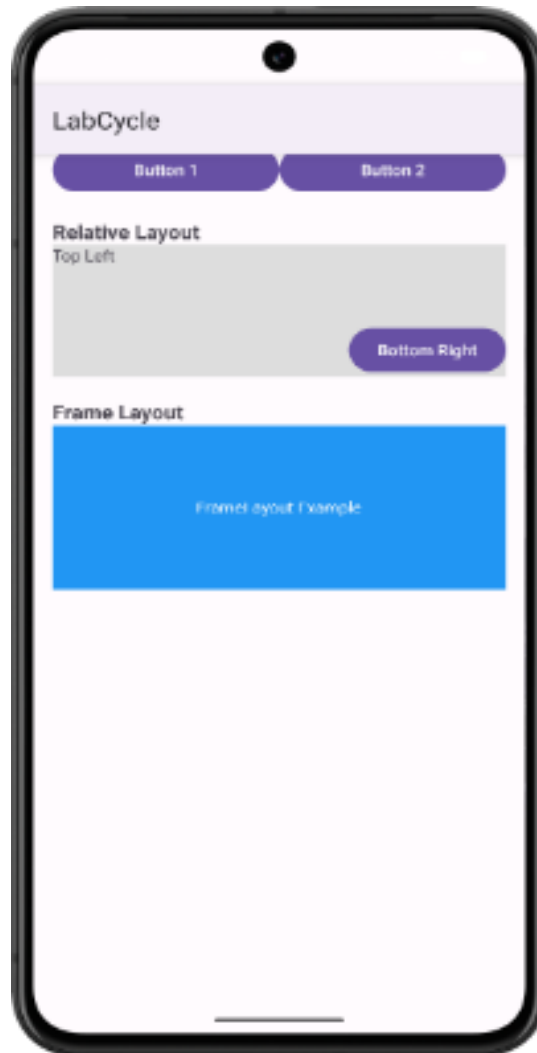
    <!-- FrameLayout -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Frame Layout"
        android:textStyle="bold"
        android:textSize="18sp"/>

    <FrameLayout
        android:layout_width="match_parent"
        android:layout_height="150dp"
        android:background="#2196F3">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="FrameLayout Example"
            android:textColor="#FFFFFF"
            android:layout_gravity="center"/>
    </FrameLayout>
</LinearLayout>
</ScrollView>

```

Output:



Program No: 4

Date: 29/06/2026

Program Title: Develop a simple calculator app.

MainActivity.java:

```
package com.example.pg04_calculator;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.widget.TextView;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {
    TextView result;
    Button ad, sub, div, mult;
    EditText no1, no2;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        result = findViewById(R.id.result);
        ad = findViewById(R.id.ad);
        sub = findViewById(R.id.sub);
        div = findViewById(R.id.div);
        mult = findViewById(R.id.mult);
        no1 = findViewById(R.id.no1);
        no2 = findViewById(R.id.no2);

        ad.setOnClickListener(v -> {
            int a = Integer.parseInt(no1.getText().toString());
            int b = Integer.parseInt(no2.getText().toString());
            int res = a + b;
            result.setText(res+"");
        });

        sub.setOnClickListener(v -> {
            int a = Integer.parseInt(no1.getText().toString());
```

```

        int b = Integer.parseInt(no2.getText().toString());
        int res = a - b;
        result.setText(res+"");
    });

    mult.setOnClickListener(v -> {
        int a = Integer.parseInt(no1.getText().toString());
        int b = Integer.parseInt(no2.getText().toString());
        int res = a * b;
        result.setText(res+"");

    });

    div.setOnClickListener(v -> {
        int a = Integer.parseInt(no1.getText().toString());
        int b = Integer.parseInt(no2.getText().toString());
        if(b!=0) {
            int res = a / b;
            result.setText(res+"");
        }
        else
            result.setText("Error: Divsion / 0");
    });
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result: "
        android:textSize="18sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"

```

```

        app:layout_constraintHorizontal_bias="0.252"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

<TextView
    android:id="@+id/result"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.52"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<EditText
    android:id="@+id/no1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Enter First Number"
    android:inputType="text"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.437"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.228" />

<EditText
    android:id="@+id/no2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Enter Second Number"
    android:inputType="text"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.437"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.355" />

```

```

<Button
    android:id="@+id/ad"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Add"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.275"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.655" />

<Button
    android:id="@+id/sub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Subtract"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.64"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.655" />

<Button
    android:id="@+id/mult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Multiply"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.275"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.78" />

<Button
    android:id="@+id/div"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Division"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"

```

</androidx.constraintlayout.widget.ConstraintLayout>

The image displays two side-by-side smartphone screens showing a calculator application. Both screens have a white background and a black status bar at the top.

Left Screen (Successful Calculation):

- Time:** 1:21
- Inputs:** The first input field contains "45" and the second input field contains "2".
- Result:** The text "Result: 47" is displayed below the input fields.
- Buttons:** There are four purple buttons with white text: "Add", "Subtract", "Multiply", and "Division".

Right Screen (Error State):

- Time:** 1:25
- Inputs:** The first input field contains "45" and the second input field contains "0".
- Result:** The text "Result: Error: Division / 0" is displayed below the input fields.
- Buttons:** There are four purple buttons with white text: "Add", "Subtract", "Multiply", and "Division".

Program No: 5

Date: 29/06/2026

Program Title: Create an app that changes text color when a button is clicked.

MainActivity.java:

```
package com.example.pg05_textcolor;

import android.graphics.Color;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.widget.TextView;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
    TextView name;
    Button color;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        color = findViewById(R.id.color);
        name = findViewById(R.id.name);
        color.setOnClickListener(v -> {
            name.setTextColor(Color.RED);
        });

    }
}
```

activity_main.xml:

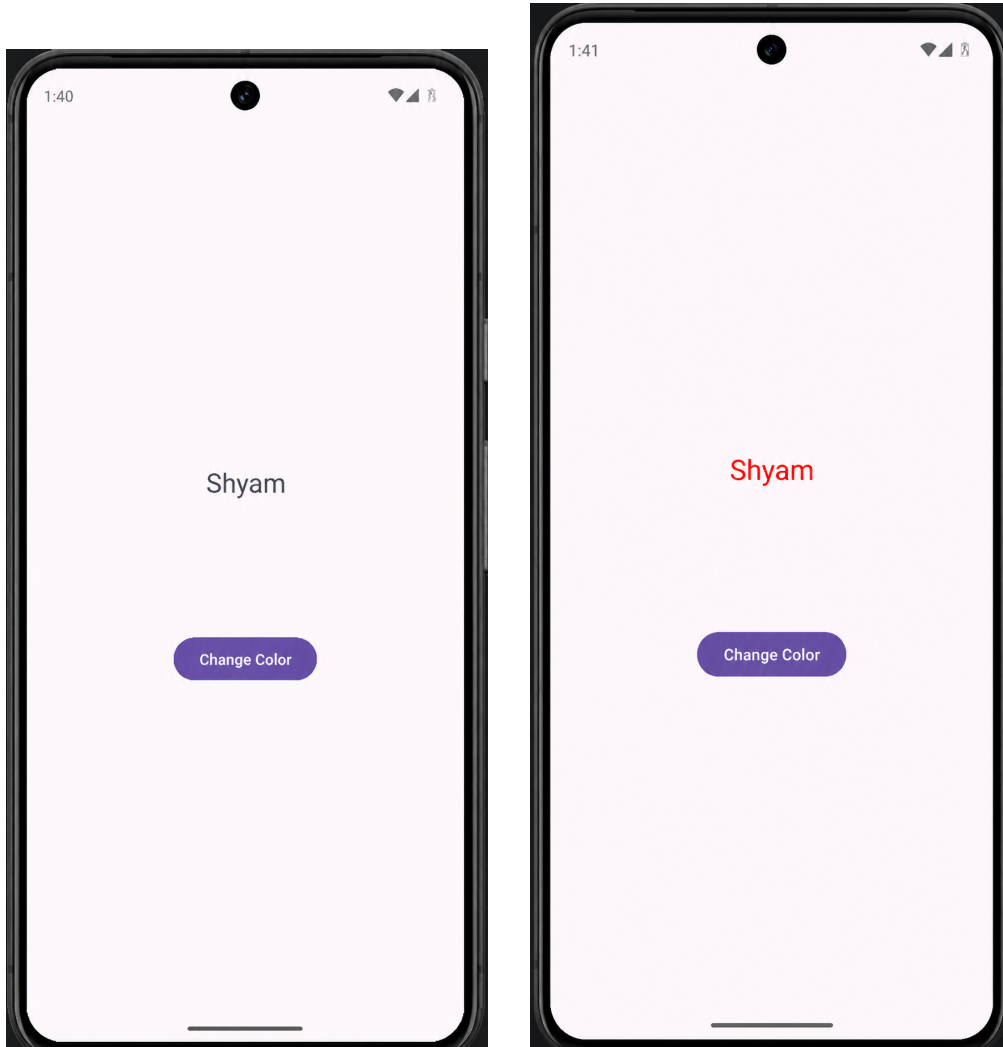
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
<TextView
    android:id="@+id/name"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Shyam"
    android:textSize="24sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.498"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.421" />

<Button
    android:id="@+id/color"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Change Color"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.613" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 6

Date: 29/06/2026

Program Title: Create an app that displays Date and Time in different formats using Linear Layout.

MainActivity.java:

```
package com.example.pg06_datetime;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.Locale;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        final TextView textView1 = findViewById(R.id.textView1);
        final TextView textView2 = findViewById(R.id.textView2);
        final TextView textView3 = findViewById(R.id.textView3);
        final TextView textView4 = findViewById(R.id.textView4);
        Button button1 = findViewById(R.id.button1);
        button1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Date currentDateTime = new Date();
                SimpleDateFormat format1 = new SimpleDateFormat("EEEE, dd
MMM yyyy", Locale.getDefault());
                SimpleDateFormat format2 = new SimpleDateFormat("MM/dd/yyyy",
Locale.getDefault());
                SimpleDateFormat format3 = new SimpleDateFormat("HH:mm:ss
z", Locale.getDefault());
                SimpleDateFormat format4 = new SimpleDateFormat("hh:mm a",
Locale.getDefault());
                textView1.setText(format1.format(currentDateTime));
                textView2.setText(format2.format(currentDateTime));
                textView3.setText(format3.format(currentDateTime));
                textView4.setText(format4.format(currentDateTime));
            }
        });
    }
}
```

```

    });
}
}

```

activity__main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

    <TextView
        android:id="@+id/textView4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button" />

```

`</LinearLayout>`

Output:

